

Marketplace Linear Search

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According to Lysecky and Vahid (2019), data structures are containers used to store various types of data, which allow specific algorithms to operate on them. Algorithms, on the other hand, are step-by-step procedures designed to achieve an expected outcome (Tutorialspoint, 2020a). When selecting a data structure, it is important to consider the operations—or algorithms—that will most frequently be performed, as the choice of data type can either enhance or hinder the performance of those algorithms (Tutorialspoint, 2020b). One such algorithm, discussed in this paper, is the linear search algorithm. In linear search, each item is sequentially compared to the target item. Once the desired item is found, it is returned (Tutorialspoint, 2020c). This algorithm is commonly used to search for items in a marketplace for purchase.

```
/Users/tea/PyCharmMiscProject/.venv/bin/python /Users/tea/Workspace/CSU-ML-AI/CSC506/1_module/linear_search.py
total distinct items in marketplace: 35
total items in stock: 1367
total items in filtered marketplace: 6
total distinct items in filtered products: 3
[MarketplaceItem(name=iPhone 16 Pro Max, color=Black, storage_gb=1000, condition=New, price=1599.0),
 MarketplaceItem(name=iPhone 16 Pro Max, color=Black, storage_gb=512, condition=New, price=1399.0),
 MarketplaceItem(name=iPhone 15 Pro Max, color=Black, storage_gb=1000, condition=New, price=1399.0)]
```

Figure 1: Market place searched.

The Algorithm In the Market Place

Linear search is typically considered inefficient, as it has a time complexity of $O(n)$, meaning the time required grows linearly with the number of items in the store. In this context, a derived data type such as `List[MarketplaceItem]` is used, which may initially seem to increase inefficiency. However, improvements can be made by applying filters to reduce the number of items examined during each search. For instance, items that are not available for sale can be

excluded from the searchable list. Additional filters—such as color or category—can further refine the results.

In a marketplace setting, searches often return multiple items that match a given criterion rather than a single result. As shown in Figure 1, most searches aim to retrieve a subset of items that meet specific conditions. In these cases, the search typically operates in the worst-case scenario of $O(\text{len}(\text{items}))$. Thus, pre-filtering data significantly improves performance by reducing the number of items processed during the linear search.

Conclusion

Although linear search is not the most efficient algorithm in general, it can be highly effective in marketplace applications when combined with appropriate filtering mechanisms. This paper has explored how filtering by attributes such as availability and color can enhance search efficiency. Other filters, including price range or brand, can further optimize search performance. When implemented thoughtfully, linear search—with targeted filtering—offers a practical solution for navigating large online stores.

References

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